BobSim

SteadyStateEval Vehicle Characterization

Measured-a_y ramp-steer isolines with robust fits

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Run Notes

- Open-loop ramp-steer at 12.5, 15, 17.5, and 20 m/s
- steerStart = 2.0 s; trim all pre-ramp samples
- Measured a_y is the analysis axis; commanded a_y only sets the ramp
- |measured a_y| < 1.0 m/s^2 is excluded; linear band = 1.0 to 4.0 m/s^2
- Understeer gradient = mean d(delta_excess)/d(a_y) over the linear band
- Response curves use split Hampel-filtered spline fits; sensitivities trim the outer 20% of |measured a_y|
- steerExcess = avgSteerAngle - wheelbase * curvature
- Tire/config: MF52, baseline_v3; Backend: OpenModelica

SteadyStateEval Summary 12.5 m/s

Metric	**Value**	**Units**
Measured a_y Range	$-17.18 \rightarrow 17.57$	$\mathrm{m/s^2}$
Roadwheel Angle Gradient	0.01026	$\frac{\mathrm{rad}}{\mathrm{m/s^2}}$
	5.76	$\frac{\mathrm{deg}}{g}$
Handwheel Angle Gradient	0.04223	$\frac{\mathrm{rad}}{\mathrm{m/s^2}}$
	23.73	$\frac{\mathrm{deg}}{g}$
Sideslip Gradient	0.00211	$\frac{\mathrm{rad}}{\mathrm{m/s^2}}$
	1.18	$\frac{\mathrm{deg}}{g}$
Understeer Gradient	0.00055	$\frac{\mathrm{rad}}{\mathrm{m/s^2}}$
	0.31	$\frac{\mathrm{deg}}{g}$
Roll Gradient	1.024	$\frac{\mathrm{deg}}{g}$
Handwheel Torque Min	-19.7	$\mathrm{N \cdot m}$
Handwheel Torque Max	19.4	$\mathrm{N \cdot m}$

SteadyStateEval Summary 15.0 m/s

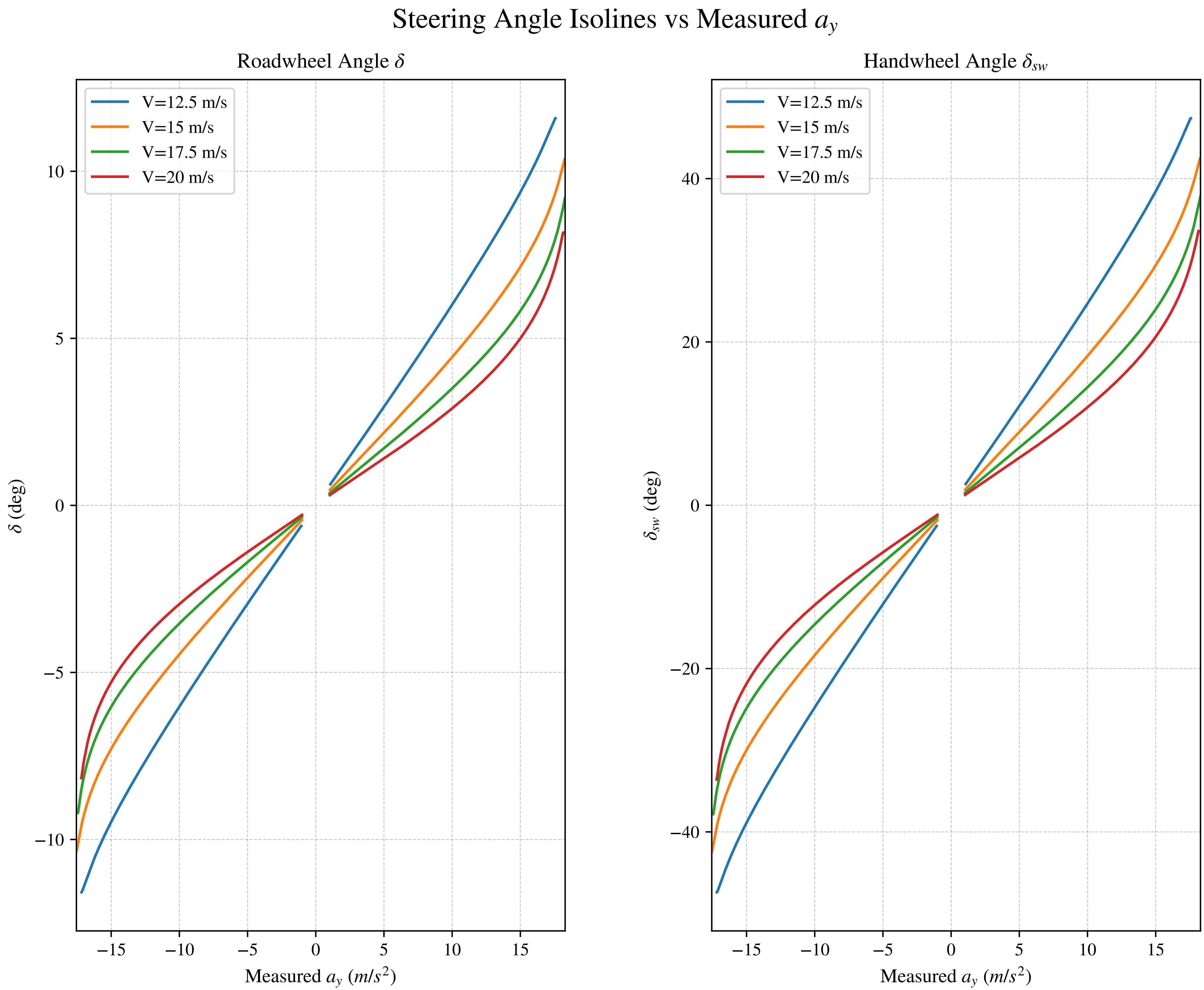
Metric	Value	Units
Measured a_y Range	$-17.55 \rightarrow 18.25$	$\mathrm{m/s^2}$
Roadwheel Angle Gradient	0.00752	$\frac{\mathrm{rad}}{\mathrm{m/s^2}}$
	4.22	$\frac{\mathrm{deg}}{g}$
Handwheel Angle Gradient	0.03099	$\frac{\mathrm{rad}}{\mathrm{m/s^2}}$
	17.42	$\frac{\mathrm{deg}}{g}$
Sideslip Gradient	0.00082	$\frac{\mathrm{rad}}{\mathrm{m/s^2}}$
	0.46	$\frac{\mathrm{deg}}{g}$
Understeer Gradient	0.00069	$\frac{\mathrm{rad}}{\mathrm{m/s^2}}$
	0.39	$\frac{\mathrm{deg}}{g}$
Roll Gradient	1.044	$\frac{\mathrm{deg}}{g}$
Handwheel Torque Min	-18.9	$\mathrm{N \cdot m}$
Handwheel Torque Max	18.6	$\mathrm{N \cdot m}$

SteadyStateEval Summary 17.5 m/s

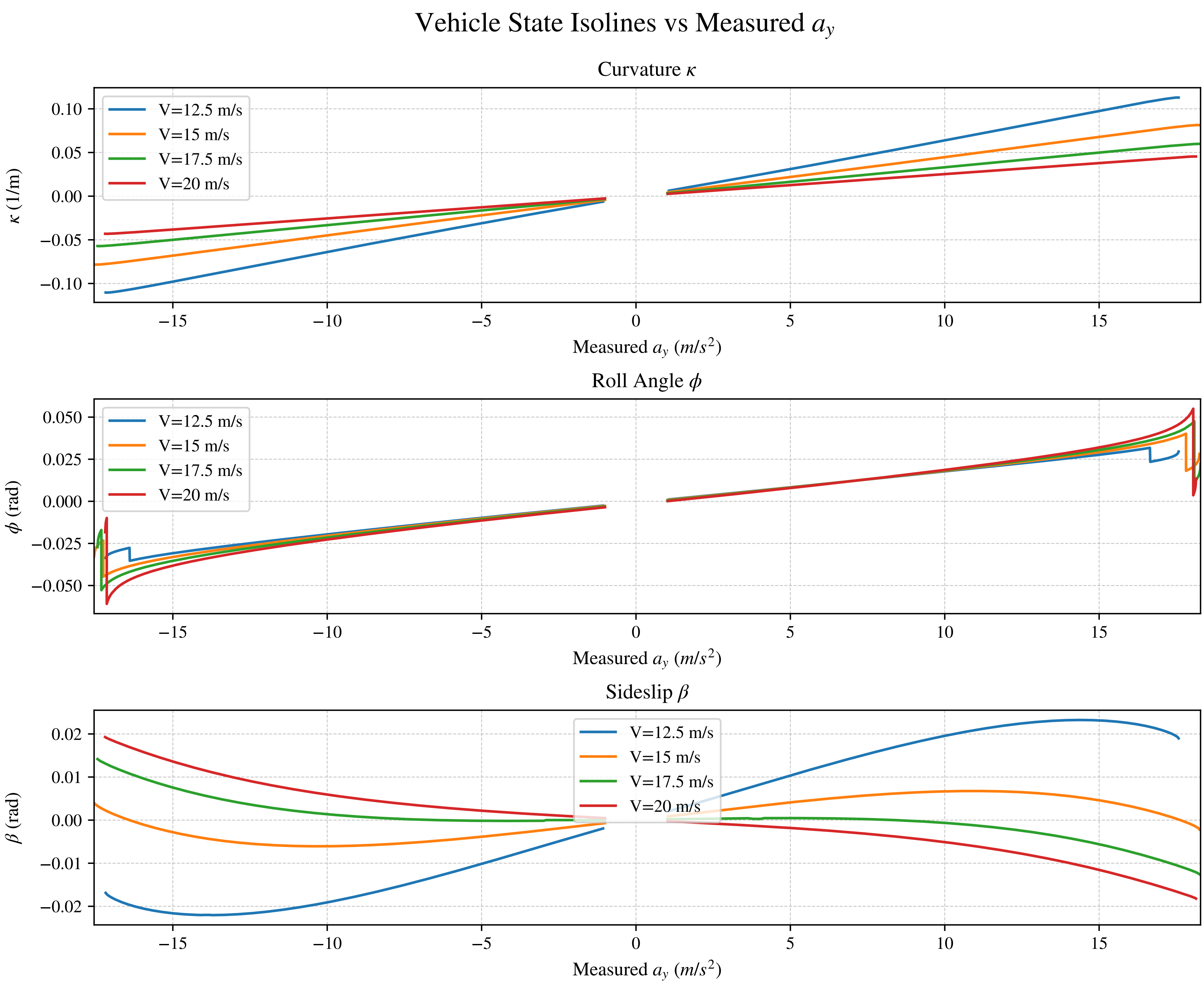
Metric	Value	Units
Measured a_y Range	$-17.44 \rightarrow 18.28$	$\mathrm{m/s^2}$
Roadwheel Angle Gradient	0.00587	$\frac{\mathrm{rad}}{\mathrm{m/s^2}}$
	3.30	$\frac{\mathrm{deg}}{g}$
Handwheel Angle Gradient	0.02421	$\frac{\mathrm{rad}}{\mathrm{m/s^2}}$
	13.61	$\frac{\mathrm{deg}}{g}$
Sideslip Gradient	0.00008	$\frac{\mathrm{rad}}{\mathrm{m/s^2}}$
	0.04	$\frac{\mathrm{deg}}{g}$
Understeer Gradient	0.00080	$\frac{\mathrm{rad}}{\mathrm{m/s^2}}$
	0.45	$\frac{\mathrm{deg}}{g}$
Roll Gradient	1.067	$\frac{\mathrm{deg}}{g}$
Handwheel Torque Min	-18.4	$\mathrm{N \cdot m}$
Handwheel Torque Max	18.1	$\mathrm{N \cdot m}$

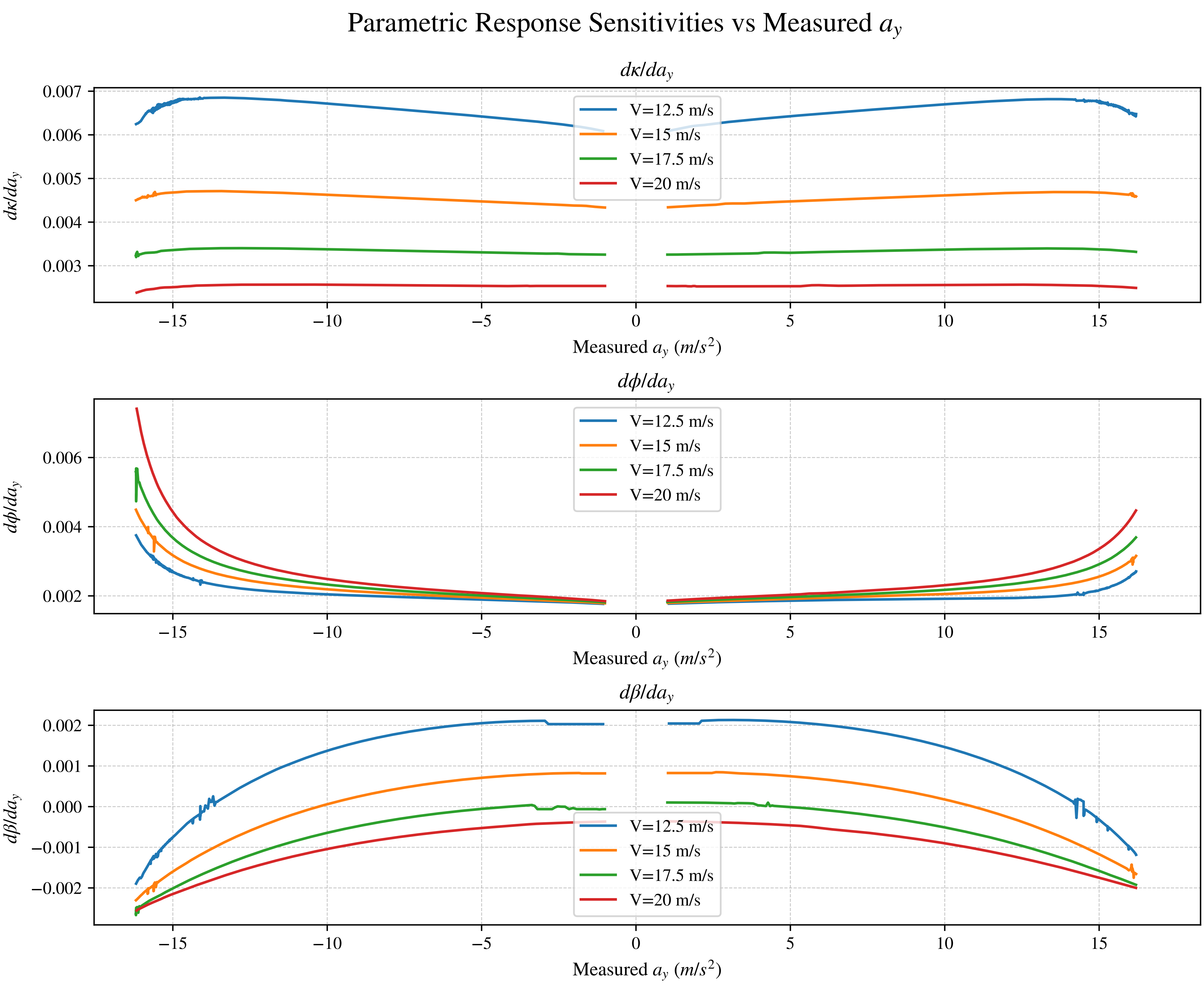
SteadyStateEval Summary 20.0 m/s

Metric	Value	Units
Measured a_y Range	$-17.19 \rightarrow 18.14$	$\mathrm{m/s^2}$
Roadwheel Angle Gradient	0.00480	$\frac{\mathrm{rad}}{\mathrm{m/s^2}}$
	2.70	$\frac{\mathrm{deg}}{g}$
Handwheel Angle Gradient	0.01983	$\frac{\mathrm{rad}}{\mathrm{m/s^2}}$
	11.15	$\frac{\mathrm{deg}}{g}$
Sideslip Gradient	-0.00039	$\frac{\mathrm{rad}}{\mathrm{m/s^2}}$
	-0.22	$\frac{\mathrm{deg}}{g}$
Understeer Gradient	0.00088	$\frac{\mathrm{rad}}{\mathrm{m/s^2}}$
	0.49	$\frac{\mathrm{deg}}{g}$
Roll Gradient	1.090	$\frac{\mathrm{deg}}{g}$
Handwheel Torque Min	-18.0	$\mathrm{N \cdot m}$
Handwheel Torque Max	17.8	$\mathrm{N \cdot m}$

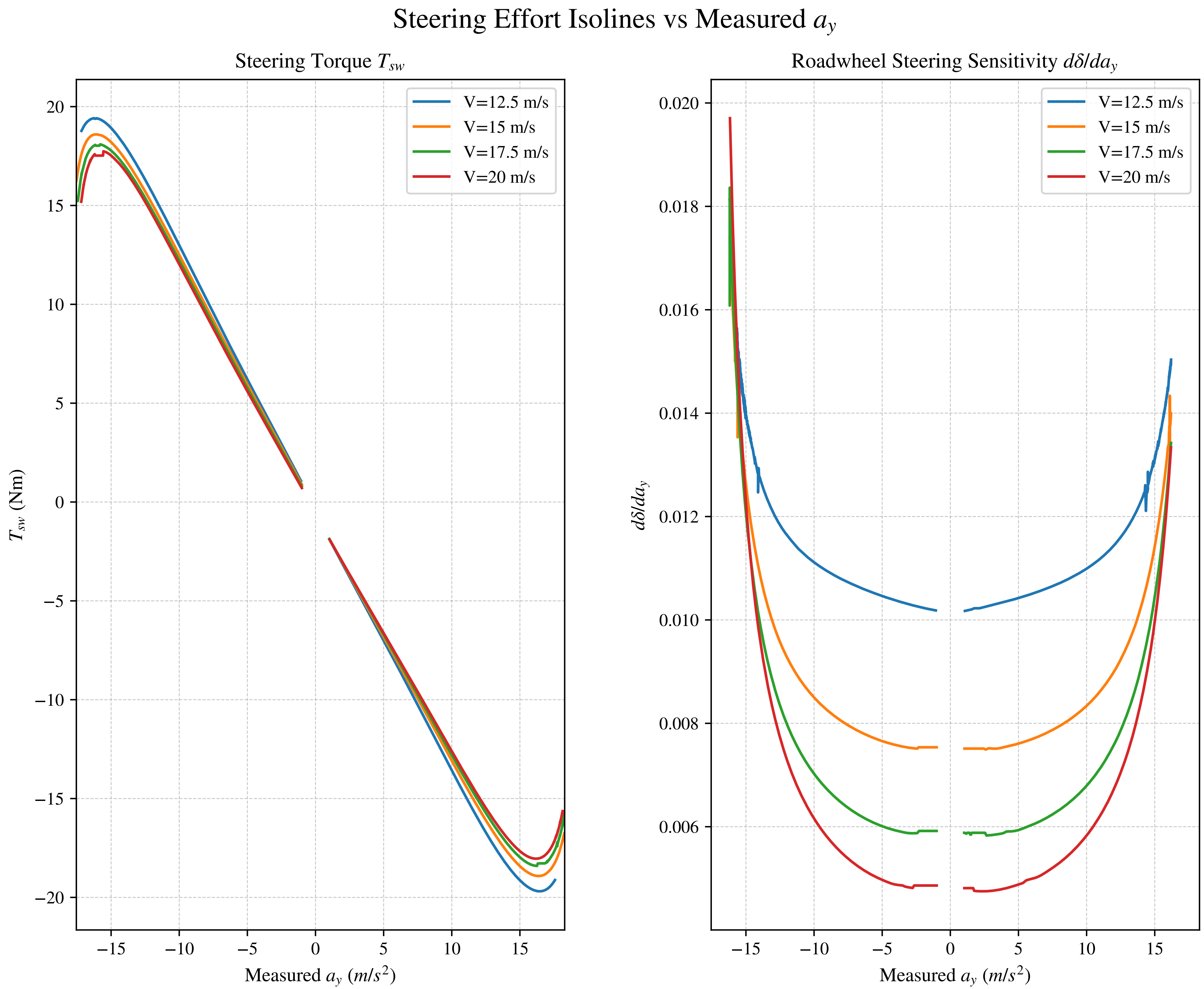


Steering Angle Isolines vs Measured a_y
Roadwheel Angle δ
Handwheel Angle δ_{sw}
V=12.5 m/s
V=15 m/s
V=17.5 m/s
V=20 m/s
δ (deg)
δ_{sw} (deg)
Measured a_y (m/s^2)
−15
−10
−5
0
5
10
15
−40
−20
20
40

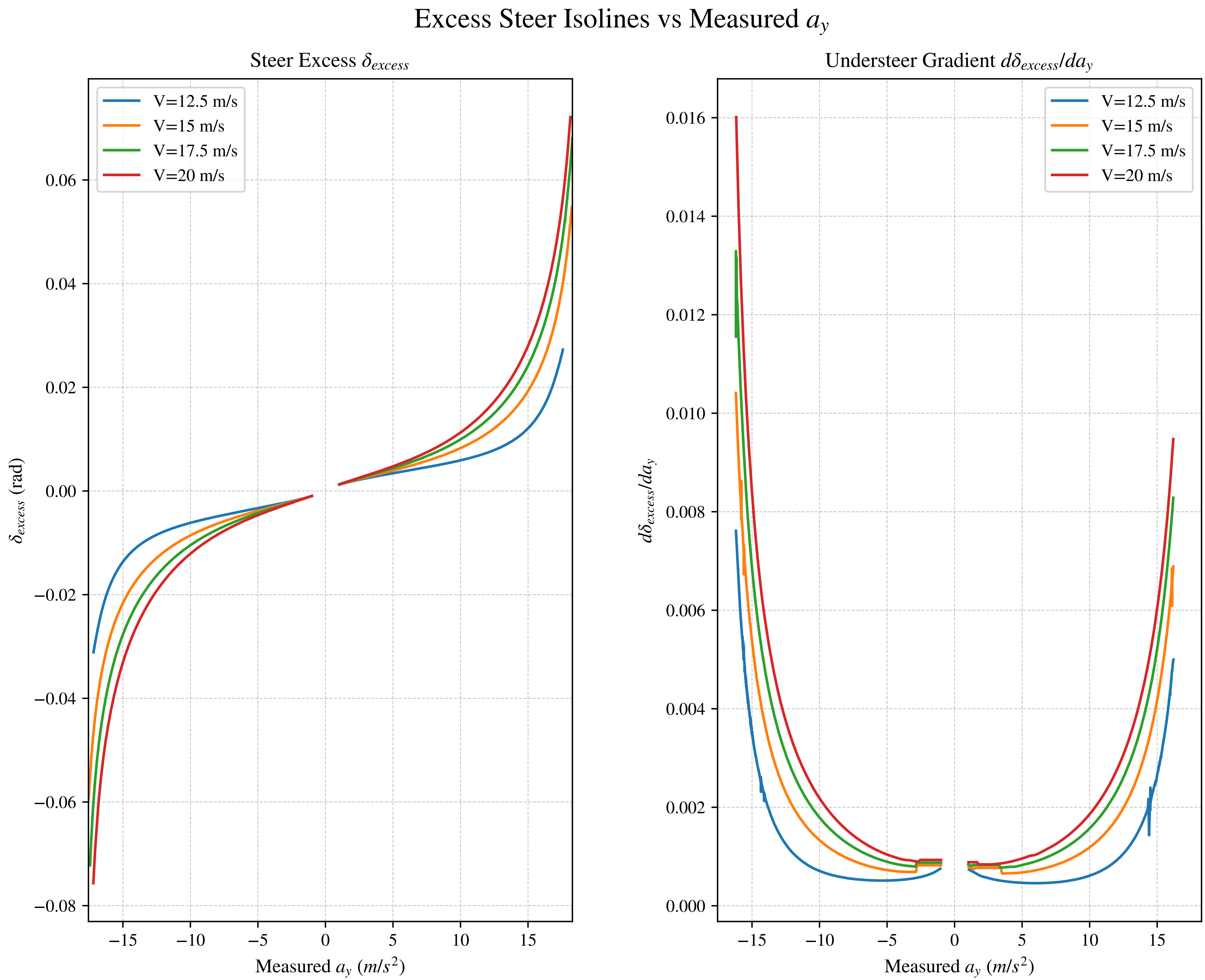

Vehicle State Isolines vs Measured a_y
Curvature κ
κ (1/m)
Measured a_y (m/s^2)
V=12.5 m/s
V=15 m/s
V=17.5 m/s
V=20 m/s
Roll Angle ϕ
ϕ (rad)
Measured a_y (m/s^2)
Sideslip β
β (rad)
Measured a_y (m/s^2)

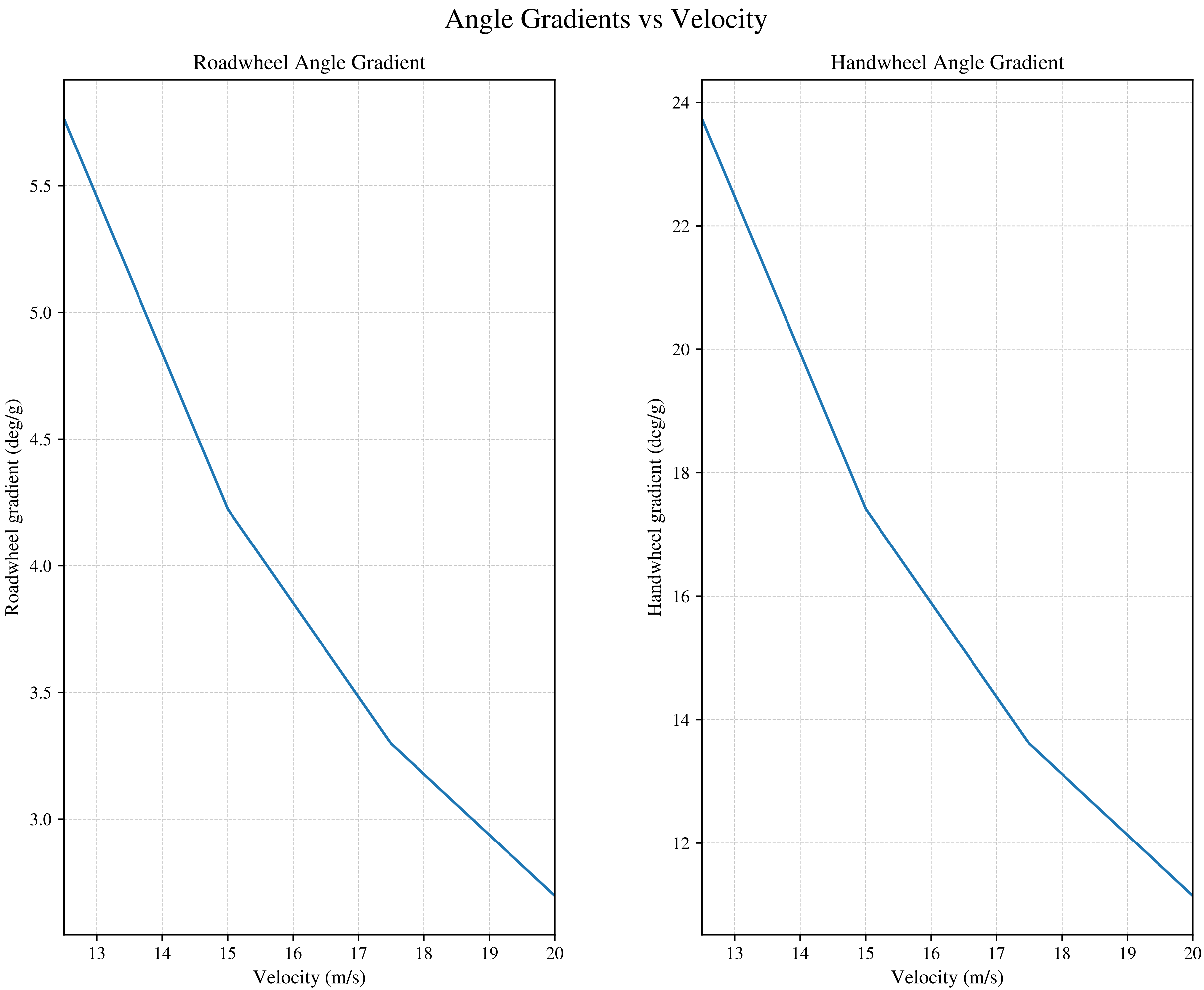


Parametric Response Sensitivities vs Measured a_y
$d\kappa/da_y$
V=12.5 m/s
V=15 m/s
V=17.5 m/s
V=20 m/s
0.007
0.006
0.005
0.004
0.003
−15
−10
−5
0
5
10
15
Measured a_y (m/s^2)
$d\phi/da_y$
0.006
0.004
0.002
$d\beta/da_y$
0.002
0.001
0.000
−0.001
−0.002

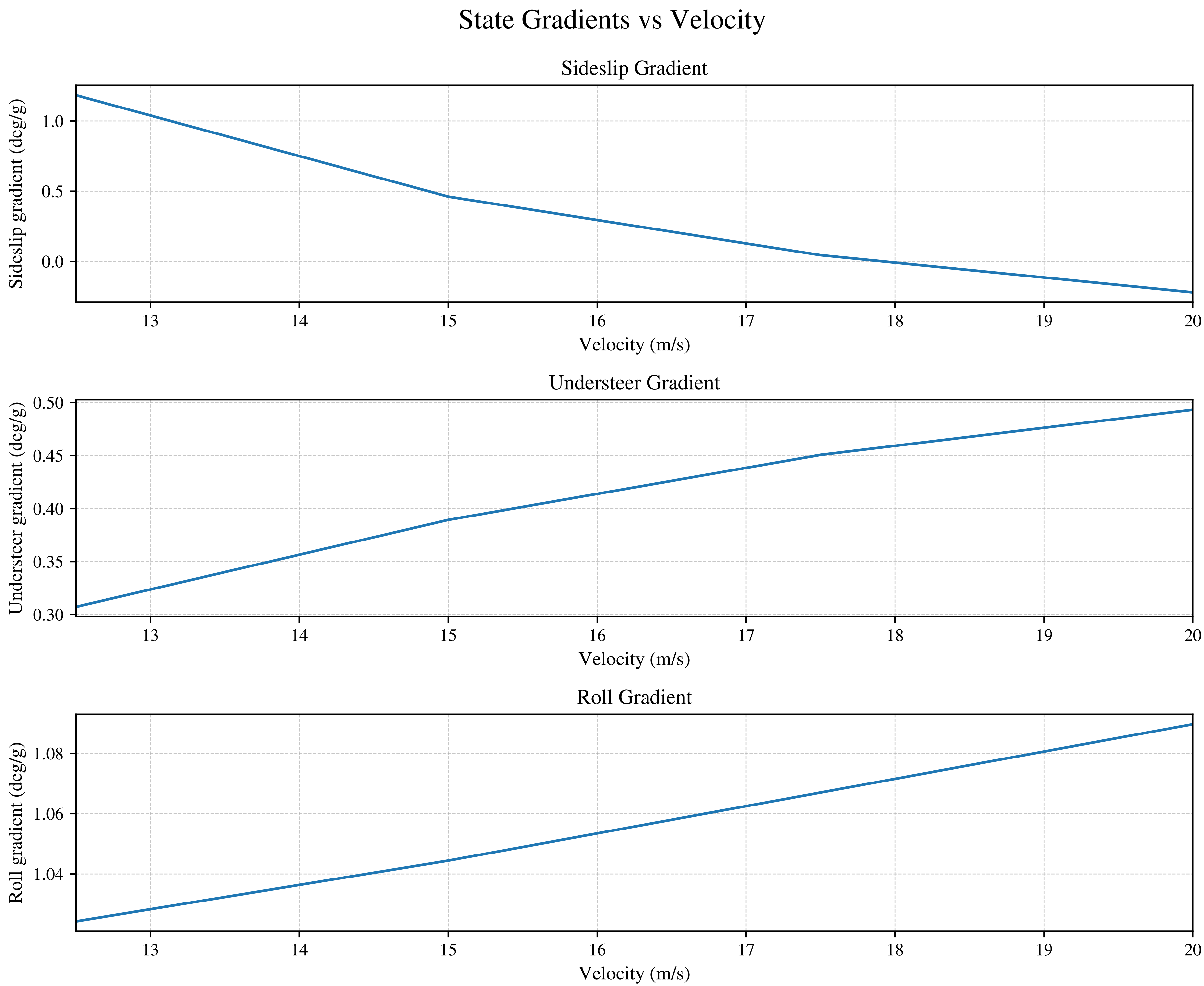


Steering Effort Isolines vs Measured a_y
Steering Torque T_{sw}
T_{sw} (Nm)
Measured a_y (m/s^2)
V=12.5 m/s
V=15 m/s
V=17.5 m/s
V=20 m/s
Roadwheel Steering Sensitivity $d\delta/da_y$
$d\delta/da_y$
Measured a_y (m/s^2)
V=12.5 m/s
V=15 m/s
V=17.5 m/s
V=20 m/s


Excess Steer Isolines vs Measured a_y
Steer Excess δ_{excess}
V=12.5 m/s
V=15 m/s
V=17.5 m/s
V=20 m/s
δ_{excess} (rad)
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
−15
−10
−5
0
5
10
15
Measured a_y (m/s^2)
Understeer Gradient $d\delta_{excess}/da_y$
V=12.5 m/s
V=15 m/s
V=17.5 m/s
V=20 m/s
$d\delta_{excess}/da_y$
0.016
0.014
0.012
0.010
0.008
0.006
0.004
0.002
0.000
−15
−10
−5
0
5
10
15
Measured a_y (m/s^2)



Angle Gradients vs Velocity
Roadwheel Angle Gradient
Roadwheel gradient (deg/g)
5.5
5.0
4.5
4.0
3.5
3.0
13
14
15
16
17
18
19
20
Velocity (m/s)
Handwheel Angle Gradient
Handwheel gradient (deg/g)
24
22
20
18
16
14
12
13
14
15
16
17
18
19
20
Velocity (m/s)



State Gradients vs Velocity
Sideslip Gradient
Sideslip gradient (deg/g)
1.0
0.5
0.0
13
14
15
16
17
18
19
20
Velocity (m/s)
Understeer Gradient
Understeer gradient (deg/g)
0.50
0.45
0.40
0.35
0.30
13
14
15
16
17
18
19
20
Velocity (m/s)
Roll Gradient
Roll gradient (deg/g)
1.08
1.06
1.04
13
14
15
16
17
18
19
20
Velocity (m/s)